

Cricket Player Performance Tracking Application :

CRICKSTATS

2. REQUIREMENT ANALYSIS

2.1 Project Overview

CRICKSTATS is a cricket player performance tracking application designed to record, manage, and analyze player statistics. It allows users to store match details such as runs, wickets, catches, strike rate, economy rate, and other performance information. The application provides performance history and reports to help players and coaches understand progress and identify areas for improvement.

2.2 Functional Requirements

- User registration and login.
- Add and manage player details.
- Add match and performance records.
- Record batting, bowling, and fielding statistics.
- Automatically calculate performance statistics.
- View previous match records.
- Search and update player information.
- Display performance reports and graphs.
- Compare player performance over different matches.
- Admin can manage users and player data.

2.3 Non-Functional Requirements

Usability: The application should be simple and easy to use.

Performance: It should provide results and reports quickly.

Security: User and player information should be protected.

Reliability: The application should store data accurately.

Scalability: It should support more players and matches in the future.

Maintainability: The system should be easy to update and maintain.

Availability: Users should be able to access their performance records whenever required.

2.4 Data Flow Diagram (Logical Flow)

User → Login → Enter Player/Match Data → Process Statistics → Store in Database → Generate Reports → User Views Performance

2.5 Technology Stack

Component	Technology
Front End	HTML, CSS, JavaScript
Back End	Python / Flask
Database	MySQL
Data Analysis	Python, Pandas
Charts	Chart.js / Matplotlib
Development Tool	Visual Studio Code
Browser	Google Chrome / Edge

2.6 Main Data Entities

User – User ID, name, email, password, role.

Player – Player ID, name, age, team, role.

Match – Match ID, date, opponent, venue, result.

Batting Performance – Runs, balls, fours, sixes, strike rate.

Bowling Performance – Overs, runs conceded, wickets, economy rate.

Fielding Performance – Catches, run-outs, stumpings.

Performance Report – Overall statistics, average, best performance, and progress.

2.7 Requirement Summary

CRICKSTATS requires a simple, secure, and reliable system that can manage player and match information and automatically calculate important cricket statistics. The application should provide an easy interface for recording performance data and generate useful reports and visualizations. These requirements ensure that players and coaches can conveniently track performance and make better decisions for player improvement.